

PROJECT 01 REPORT

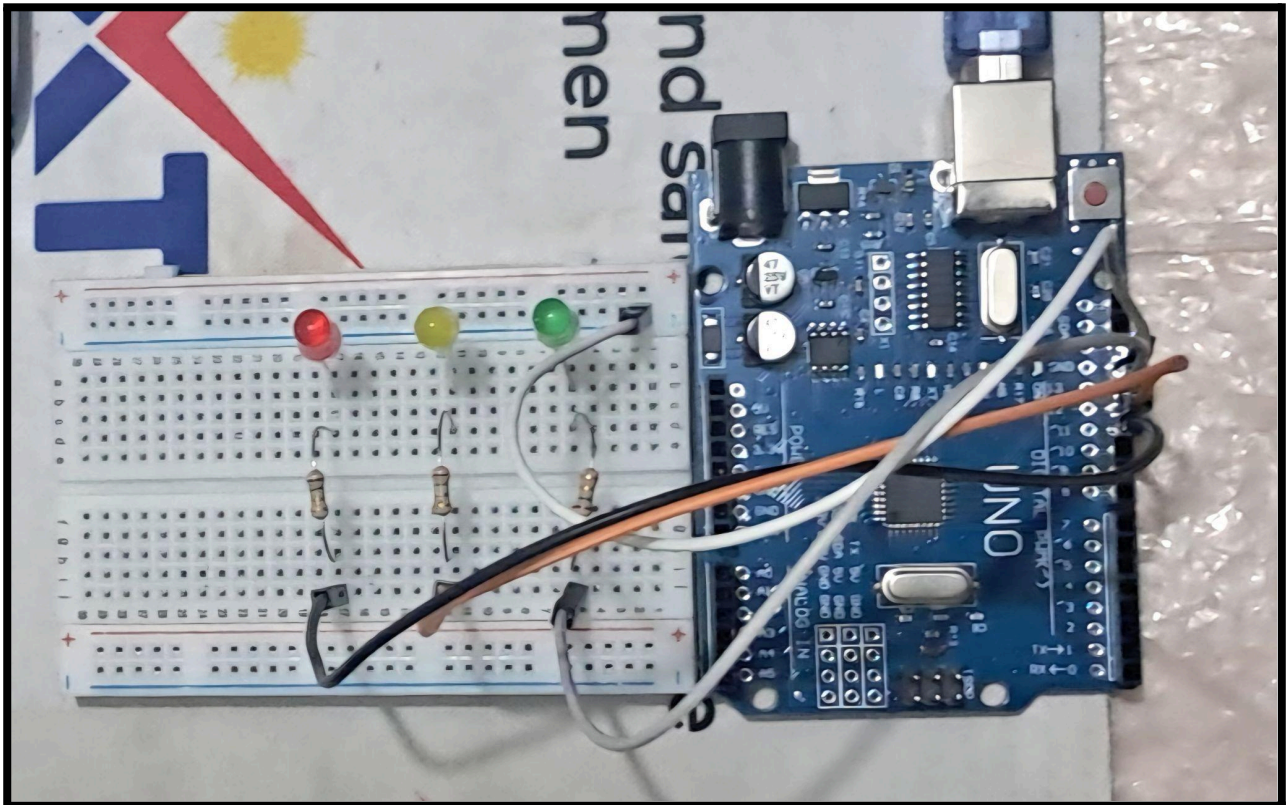
Arduino LED Control and Pattern Generation System

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Duration : 2 days



1. Objective

The objective of this project was to gain practical experience with Arduino Uno, breadboard, GPIO pins, LED interfacing, resistors and PWM generation.

The project served as the foundation for future embedded systems projects and made familiarity with hardware wiring and Arduino programming.

2. Components Used

Component	Quantity
Arduino Uno R3	1
Breadboard	1
LEDs (Red,Yellow,Green,Blue)	Multiple
Resistors(150Ω,300Ω,1kΩ)	Multiple
Jumper Wires	Multiple
USB Cable	1

3. Software Used

- Arduino IDE
- Windows Laptop

4. Theory

An LED (Light Emitting Diode) is a semiconductor device that emits light when current flows through it.

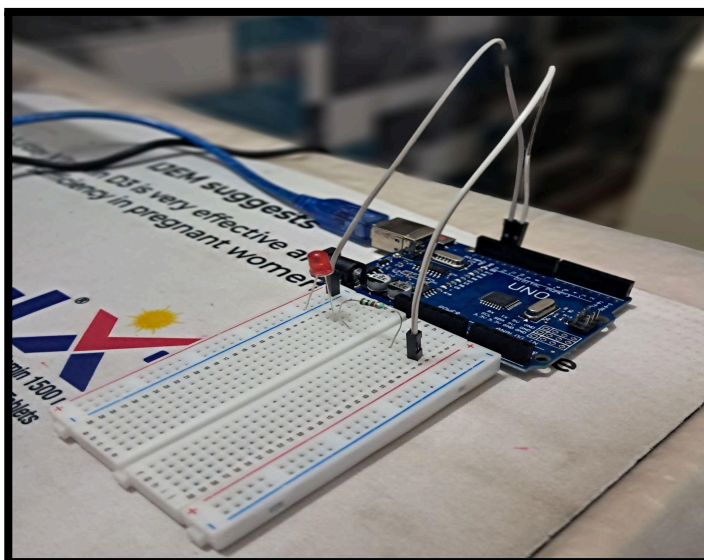
Since LEDs require limited current, resistors are connected in series to prevent the excessive flow of current.

The Arduino controls the LEDs using its digital output pins. By changing the output state between HIGH and LOW, LEDs can be turned ON and OFF.

PWM (Pulse Width Modulation) allows the control of brightness by rapidly switching the LED ON and OFF at high speed while varying the duty cycle.

5. Tasks Performed

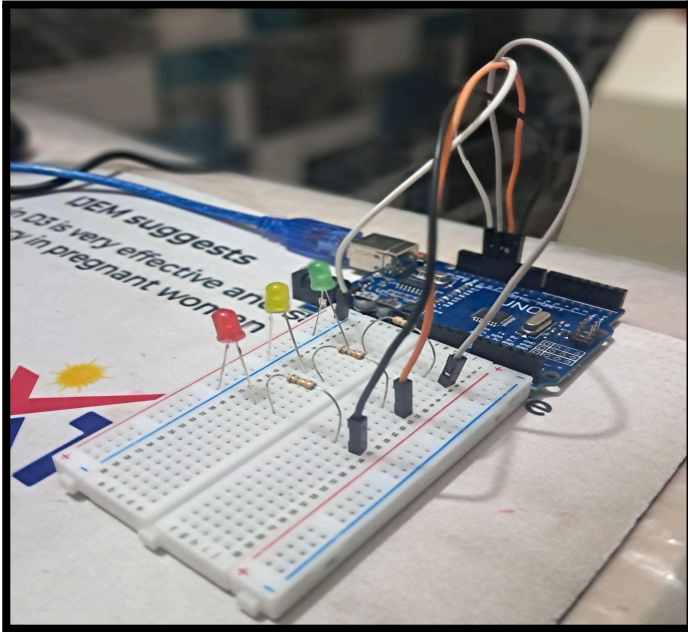
➤ Single LED Blink



Vedio

A single LED was connected to Arduino and programmed to blink periodically. (project video Demonstration

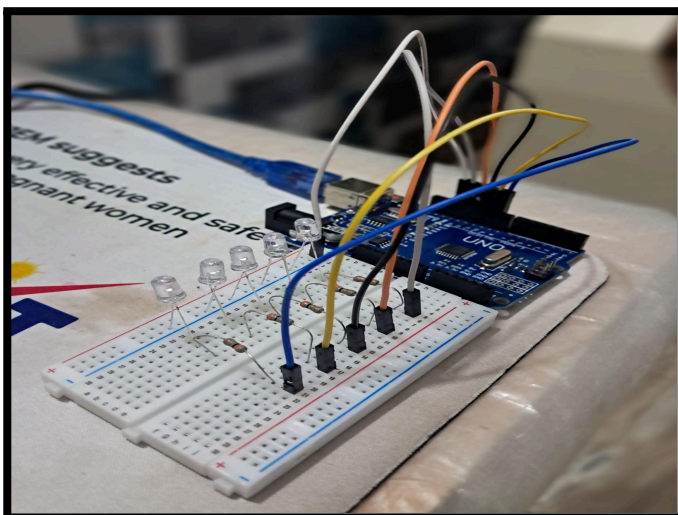
➤ Traffic Light System



Vedio

Three LEDs were connected to represent traffic signals and were programmed to turn ON sequentially according to predefined timings.

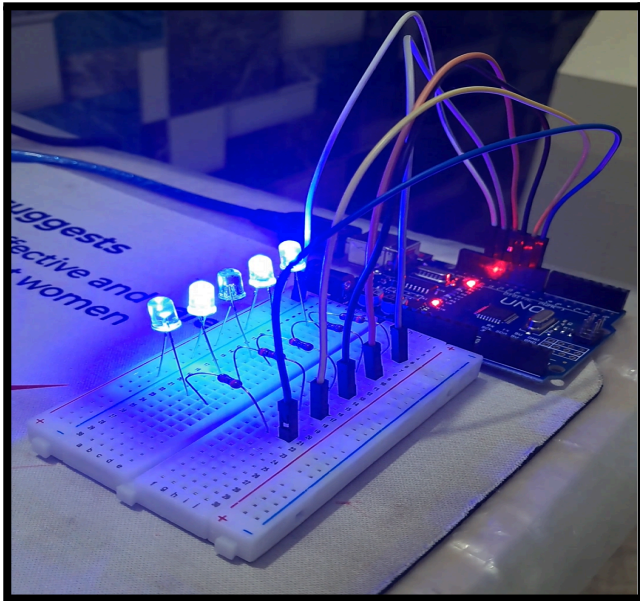
➤ LED Chaser Pattern



Vedio

LEDs were made to glow one after another continuously.

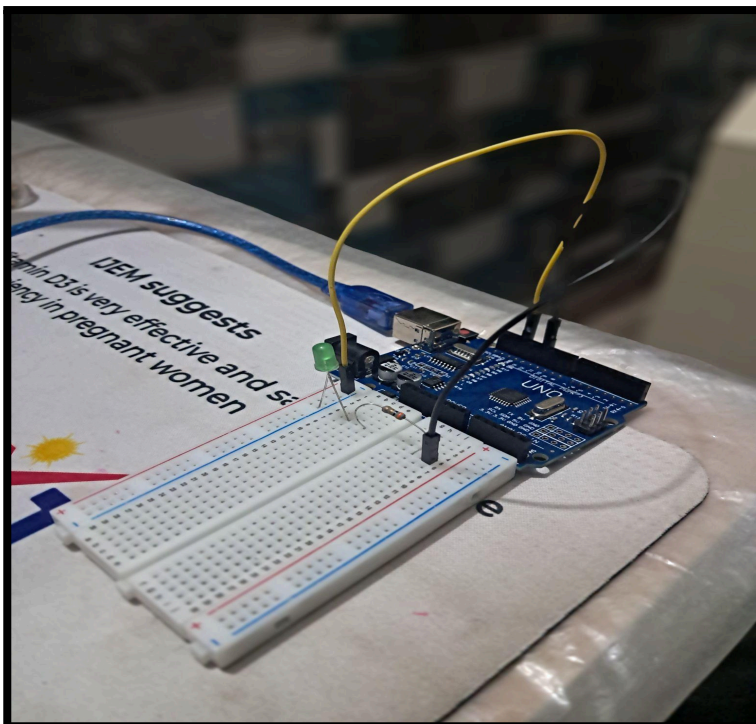
➤ Back and Forth Pattern



Vedio

LEDs were programmed to illuminate in one direction

➤ PWM Brightness Control



Vedio

PWM signals were generated using Arduino PWM pins to vary LED brightness creating a breathing effect.

6. Learning Outcomes

The following concepts were learned in making of this project:

- Arduino IDE installation and setup
- Uploading programs to Arduino Uno
- Understanding setup() and loop()
- GPIO Pin Configuration
- Breadboard Internal Connections
- LED Polarity
- Current Limiting Resistors
- Multi-LED Control
- Sequential Pattern Generation
- PWM Based Brightness Control

7. Problems Faced

- I. Initially, program upload failed because the sketch was not saved.
- II. Resistors from the resistor kit were accidentally mixed together.

8. Observations

- The built-in Arduino LED is connected to Pin 13.
- External LEDs connected to Pin 13 blink simultaneously with the onboard LED.
- $1\text{k}\Omega$ resistors provide lower brightness as compared to 150Ω and 300Ω but operate safely.
- PWM allows smooth brightness control without changing the actual output voltage.

9. Applications and Future Scope

The concepts and techniques learned in this project can be applied to various real-world systems such as traffic signal controllers, industrial status indicators, home automation systems, embedded control applications, motor speed control systems, and LED dimming circuits using PWM.

The project can be further enhanced by incorporating additional features such as push-button control, pedestrian crossing signals, sensor-based traffic management, countdown timer displays, IoT-based monitoring and control using ESP32, and custom PCB implementation for a compact and professional hardware design.

10. Conclusion

This project successfully introduced the fundamentals of embedded systems using Arduino Uno. Practical experience was gained in circuit construction, Arduino programming, digital output control, and PWM generation providing a strong foundation for future sensor-based and IoT-based embedded projects.